



United International University
Department of Computer Science and Engineering
CSE 4509: Operating Systems
Final Examination: Fall 2025
Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be
expelled from the trimester / program as per UIU
disciplinary rules.

Answer all the questions. Numbers to the right of the questions denote their marks.

1(a)	With the help of a block diagram, explain how Translation Lookahead Buffer (TLB) improves the performance of a paging system.	[4]
1(b)	In a multiprogramming system, two memory management schemes are proposed: one uses variable-sized contiguous allocation and the other uses paging. Over time, both systems begin to waste memory, but in different ways. Explain why one system primarily suffers from external fragmentation while the other primarily suffers from internal fragmentation. Why cannot a single system completely eliminate both types of fragmentation at the same time?	[4]
2(a)	A system uses demand paging where the main memory access time is 100 ns, the page fault service time for handling the interrupt and restarting the process is 8 μ s, the time to swap a page out to disk is 4 ms, and the time to swap a page into memory is 10 ms. Find the maximum value of page fault rate such that the Effective Access Time does not exceed 200 ns.	[4]
2(b)	For the following reference string of referenced page numbers, perform LRU to find out the number of page faults. Assume that maximum 3 frames can be stored in the memory for this process. 7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1	[4]
2(c)	An operating system designer suggested the separation of logical and physical address space for faster process creation. After implementing the design, he noticed that if the child process modifies a data, it is also automatically modified in the parent process, which should not happen. Explain how the separation of logical and physical address space helps to faster process creation. What can be the possible fault in the implementation? How can you solve it?	[4]

3	The UIU Smart Campus Portal runs multiple services concurrently: Course Registration (CRS), Fee Payment (FPS), and Grade Processing (GPS). Each service may require exclusive access to shared resources: Student Database (R1), Payment Gateway (R2), Notification Service (R3). Sometimes, CRS locks R1 and waits for R3, FPS locks R2 and waits for R1, and GPS locks R3 and waits for R2. Some transactions never complete. (a) Explain why this situation may lead to deadlock using the four necessary conditions and a resource allocation graph from the above description. (b) Consider a system with 5 threads (T0–T5) and 4 resource types A(11), B(12), C(16), D(16). The Allocation and Maximum demand for each process are given below: <table><tr><th rowspan="2">Process</th><th colspan="4">Allocation</th><th colspan="4">Max</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>T0</td><td>2</td><td>1</td><td>3</td><td>4</td><td>5</td><td>3</td><td>6</td><td>7</td></tr><tr><td>T1</td><td>3</td><td>1</td><td>2</td><td>1</td><td>3</td><td>4</td><td>3</td><td>5</td></tr><tr><td>T2</td><td>2</td><td>0</td><td>5</td><td>3</td><td>4</td><td>3</td><td>7</td><td>4</td></tr><tr><td>T3</td><td>3</td><td>5</td><td>4</td><td>2</td><td>9</td><td>8</td><td>6</td><td>9</td></tr><tr><td>T4</td><td>1</td><td>2</td><td>1</td><td>2</td><td>6</td><td>3</td><td>2</td><td>5</td></tr></table> Using Banker's Algorithm, determine whether the system is in a safe state. If the system is in a safe state, write down a safe sequence; otherwise, specify which thread(s) can be executed.	Process	Allocation				Max				A	B	C	D	A	B	C	D	T0	2	1	3	4	5	3	6	7	T1	3	1	2	1	3	4	3	5	T2	2	0	5	3	4	3	7	4	T3	3	5	4	2	9	8	6	9	T4	1	2	1	2	6	3	2	5	[5] [5]
Process	Allocation				Max																																																											
	A	B	C	D	A	B	C	D																																																								
T0	2	1	3	4	5	3	6	7																																																								
T1	3	1	2	1	3	4	3	5																																																								
T2	2	0	5	3	4	3	7	4																																																								
T3	3	5	4	2	9	8	6	9																																																								
T4	1	2	1	2	6	3	2	5																																																								
4	Sam is designing a file system for a university research server. The server is used by students and faculty for different purposes: • Lecture videos are very large, rarely modified once uploaded, and are usually read sequentially. • Source code files are small to medium in size and are frequently edited and accessed randomly. • Research data files grow gradually over time as new experimental results are appended. Based on this scenario, answer the following questions. (a) If Sam plans to implement contiguous allocation, linked list allocation, and i-node allocation in this scenario, choose the most suitable allocation method for each file type. (b) Which single file allocation method would you recommend for this system overall? Explain the reasoning behind your chosen method. (c) Which of these three allocation methods has the worst access performance? How can it be improved?	[6] [2] [2]																																																														